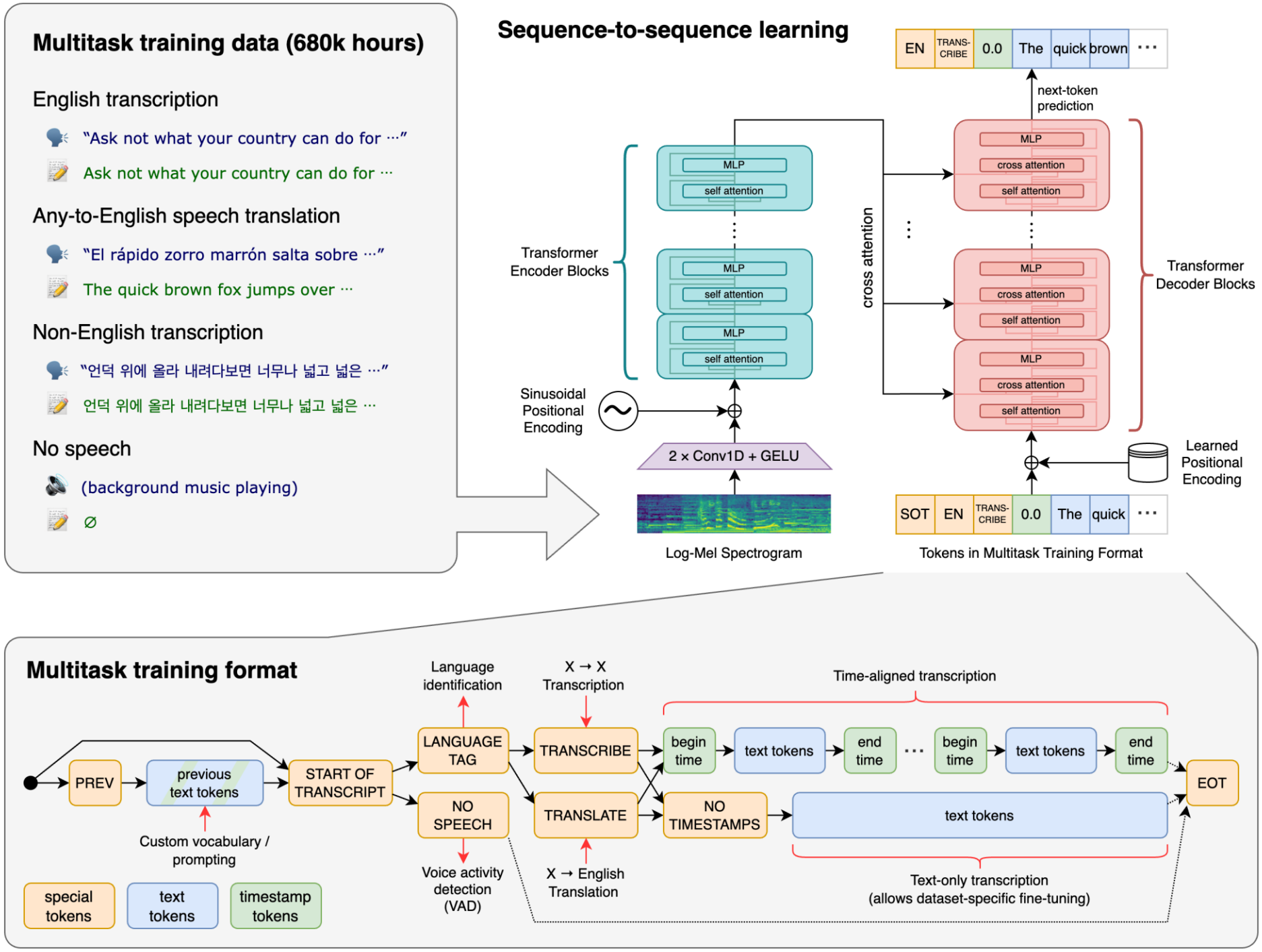
# ASR

1. Whisper 2022
2. SenseVoice Tongyi 2024.7
3. Qwen2-ASR 7B 2024.7
4. VibeVoice 7B microsoft 2025
5. VibeVoiceASR 2026.3
6. Step-Audio 2025.8
7. Mimo-Audio 7B. 2025.12
8. Moss-Transcribe-Diarize  0.9B  2026.7
9. Cohere-transcribe-03-2026
10. Qwen3-ASR-1.7B. 2026.1

| | Time | Organization | Paper | |
|---|---|---|---|---|
| Whisper<br>Whisper-turbo<br>1550M | 2022.Sep<br>2022.Dec | OpenAI | Robust Speech Recognition via Large-Scale Weak Supervision | https://arxiv.org/pdf/2212.04356 |
| SenseVoice 7B | 2024.7 | Alibaba Group Tongyi SpeechTeam | FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs | https://arxiv.org/pdf/2407.04051 |
| Qwen2-Audio 7B | 2024.7 | Alibaba | Qwen2-Audio Technical Report | https://arxiv.org/pdf/2407.10759 |
| VibeVoice 7B<br>VibeVoiceASR 7B | 2025<br>2026 3 | Microsoft | VibeVoice Technical Report<br>VIBEVOICE-ASR Technical Report | https://arxiv.org/pdf/2508.19205<br>https://arxiv.org/pdf/2601.18184 |
| Step-Audio | 2025.8 | StepFun Audio Team | Step-Audio 2 Technical Repor | https://arxiv.org/pdf/2507.16632 |
| Mimo-Audio 7B | 2025.12 | XiaoMi | MiMo-Audio: Audio Language Models are Few-Shot Learners | https://arxiv.org/pdf/2512.23808 |
| Moss-Transcribe-diarize 0.9B | 2026. 7 | MOSI.AI | MOSS Transcribe Diarize Technical Report | https://arxiv.org/pdf/2601.01554 |
| Qwen3-ASR 1.7B | 2026.1 | Qwen | Qwen3-ASR Technical Report | https://arxiv.org/pdf/2601.21337 |

Robust Speech Recognition via Large-Scale Weak Supervision - 2022



FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs Alibaba Group -07-2024

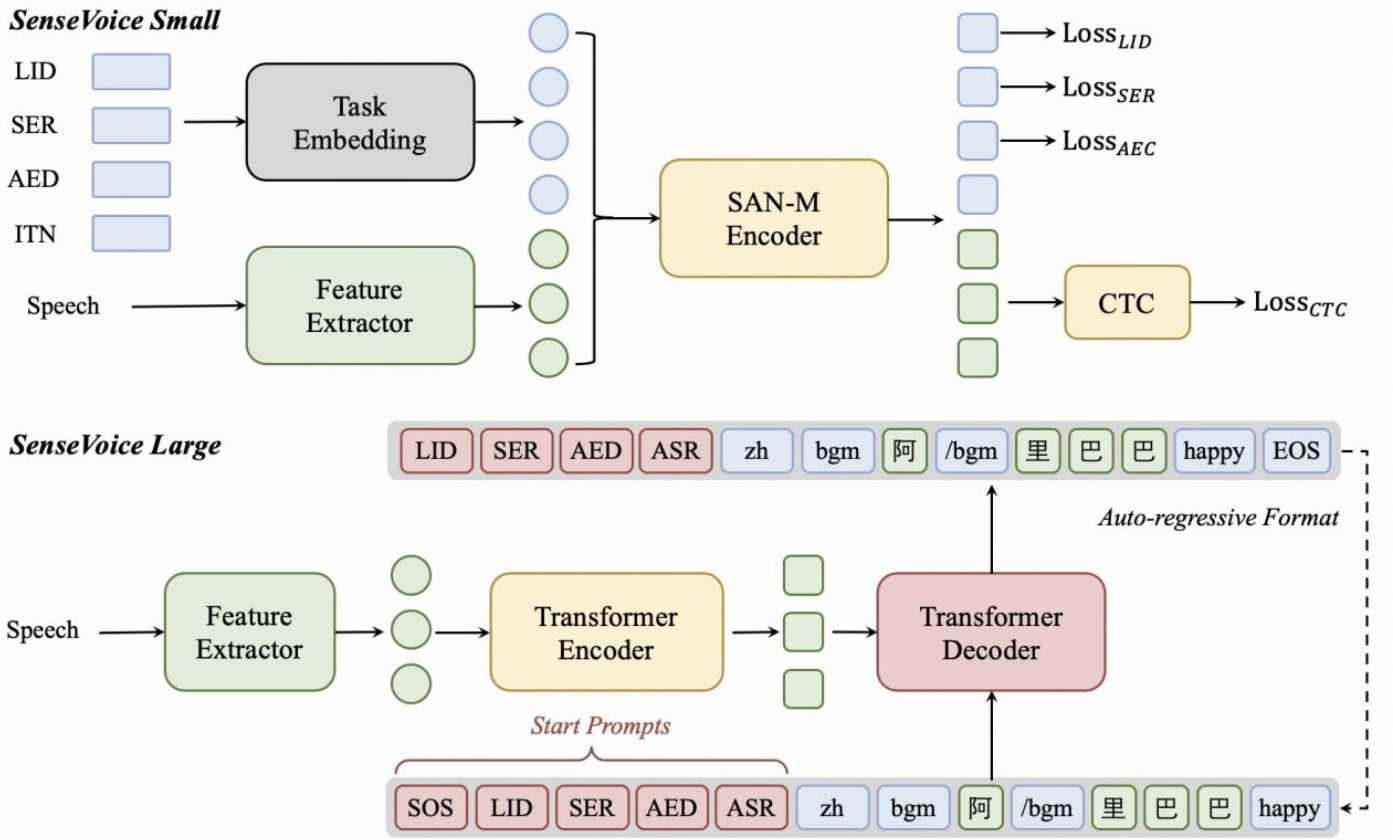



Figure 2: SenseVoice is a comprehensive speech foundation model designed to perform various speech understanding tasks, including Automatic Speech Recognition (ASR), Language Identification (LID), Speech Emotion Recognition (SER), and Audio Event Detection (AED). SenseVoice-Small [Top]: An encoder-only model optimized for rapid speech understanding. It offers high-speed processing while supporting 5 languages. SenseVoice-Large [Bottom]: An encoder-decoder model aimed at achieving more precise speech understanding across a broader range of languages. It excels in accuracy and supports an extensive set of language capabilities.

Qwen2-Audio Technical Report 2024-02-02

Qwen2-Audio-7B

1. Generalization LALM

SLU(Spoken language understanding)
SER(Speech Emotion Recognition)
ASR(Audio Speech Recognition)
S2TT(Speech-to-Text Translation)
VSC(Vocal Sound Classification)

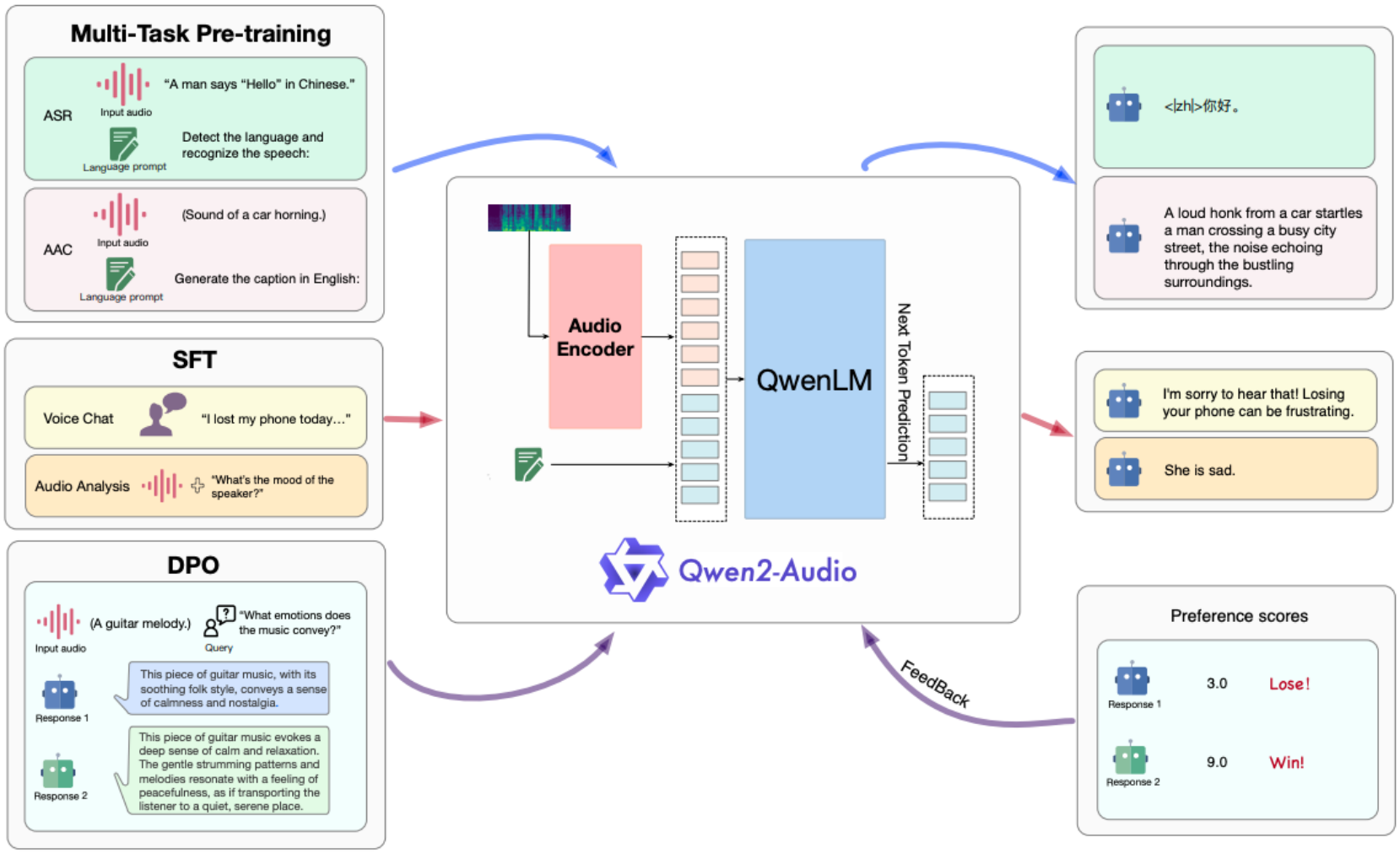



Figure 2: The overview of three-stage training process of Qwen2-Audio.

VIBEVOICE Technical Report -2025

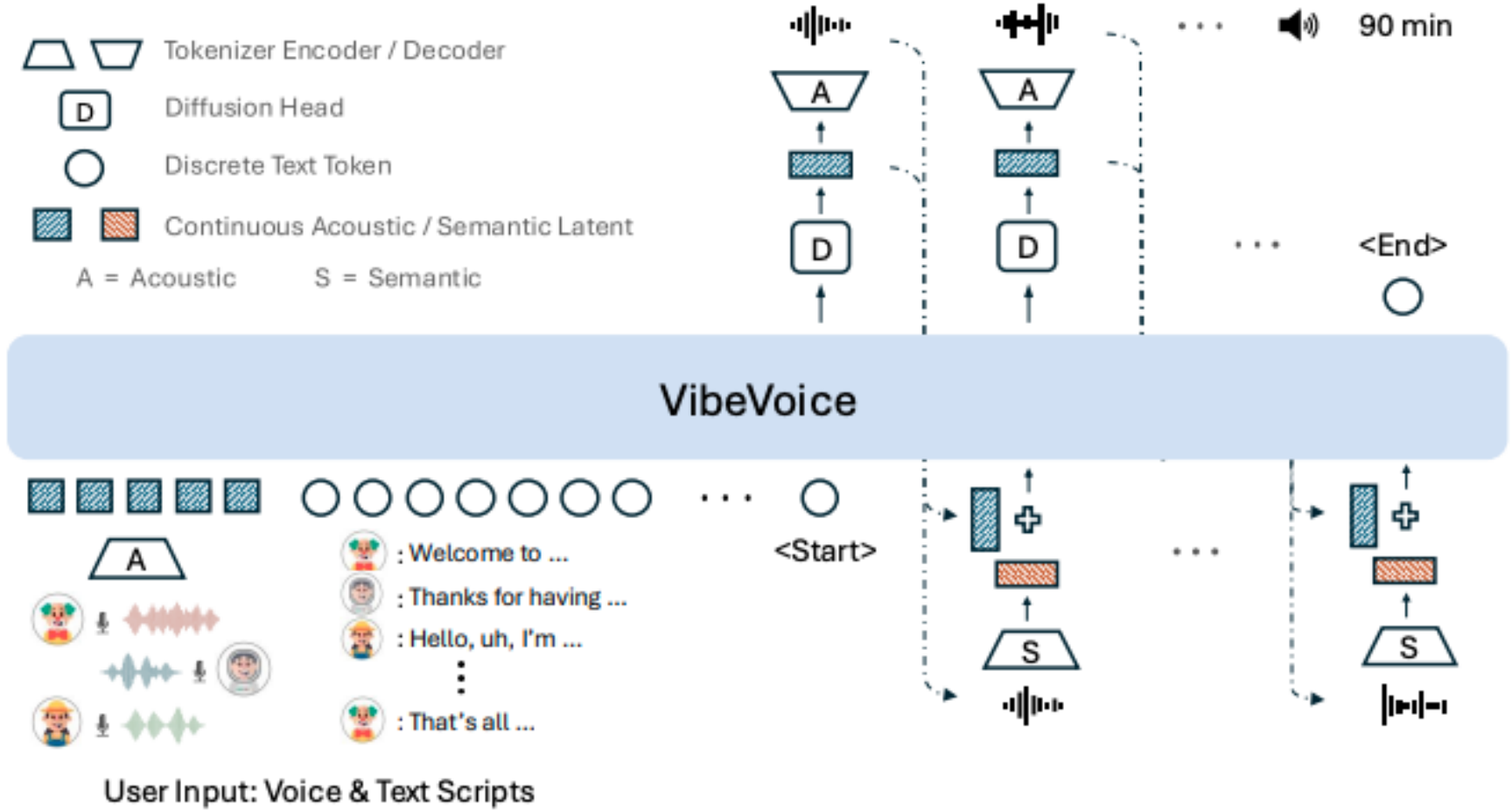



Figure 2: VIBEVOICE employs next token diffusion framework as in LatentLM [SBW+24] to synthesize long-form and multi-speaker audios. Voice prompts and text scripts provide initial input. VIBEVOICE processes hybrid context features, and its hidden states condition a token level Diffusion Head (D), which predicts acoustic VAE for speech segments, subsequently recovered by acoustic decoder (A).

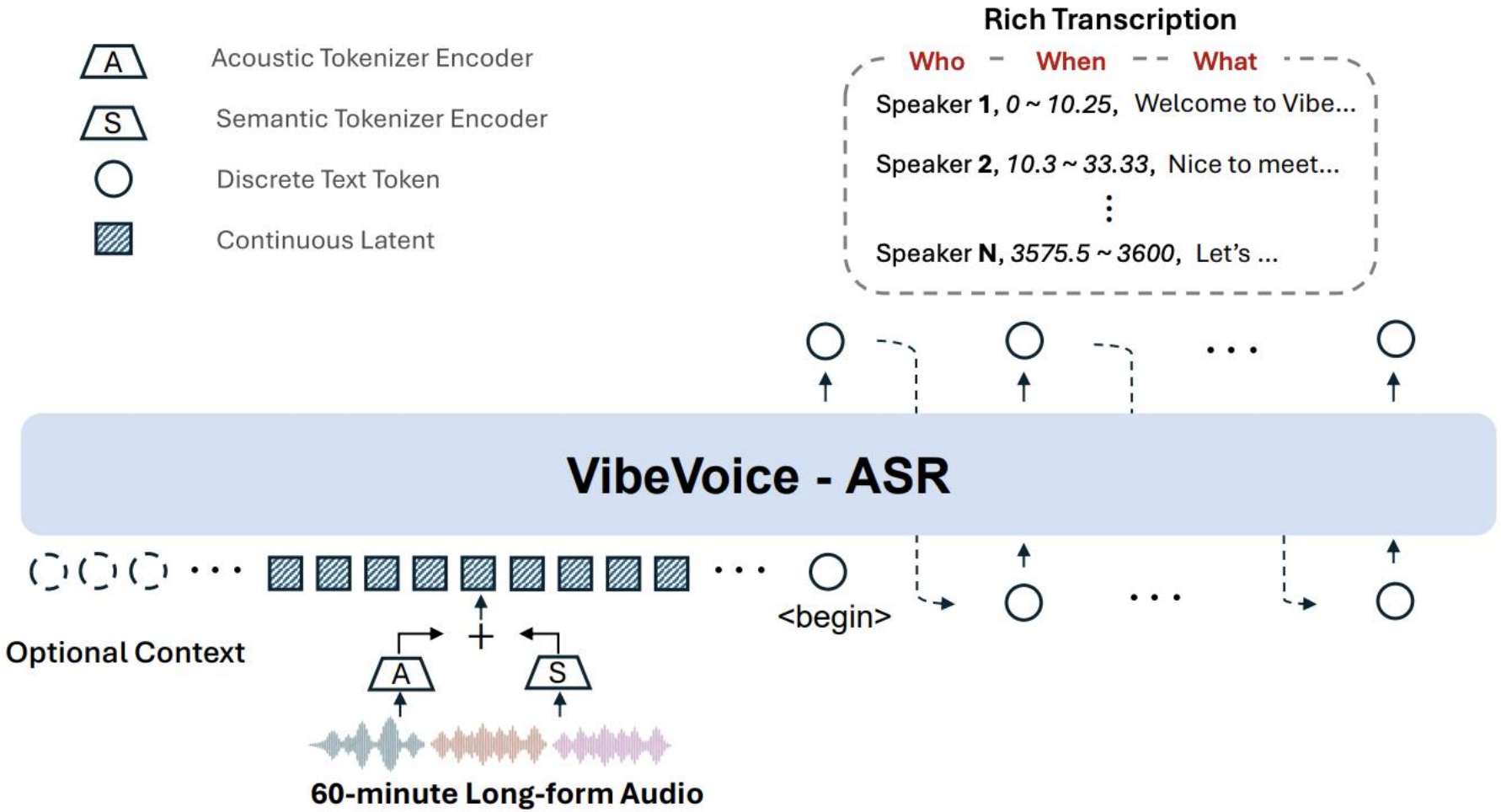



Figure 2: The architectural overview of VIBEVOICE-ASR. VIBEVOICE-ASR processes 60-minute long-form audio in a single pass by ingesting continuous latents from dual-tokenizers alongside optional user-provided context. The output is a generated stream of Rich Transcription, explicitly interleaving Speaker ID (Who), Timestamps (When), and Content (What)

MiMo-Audio: Audio Language Models are Few-Shot Learners-2025

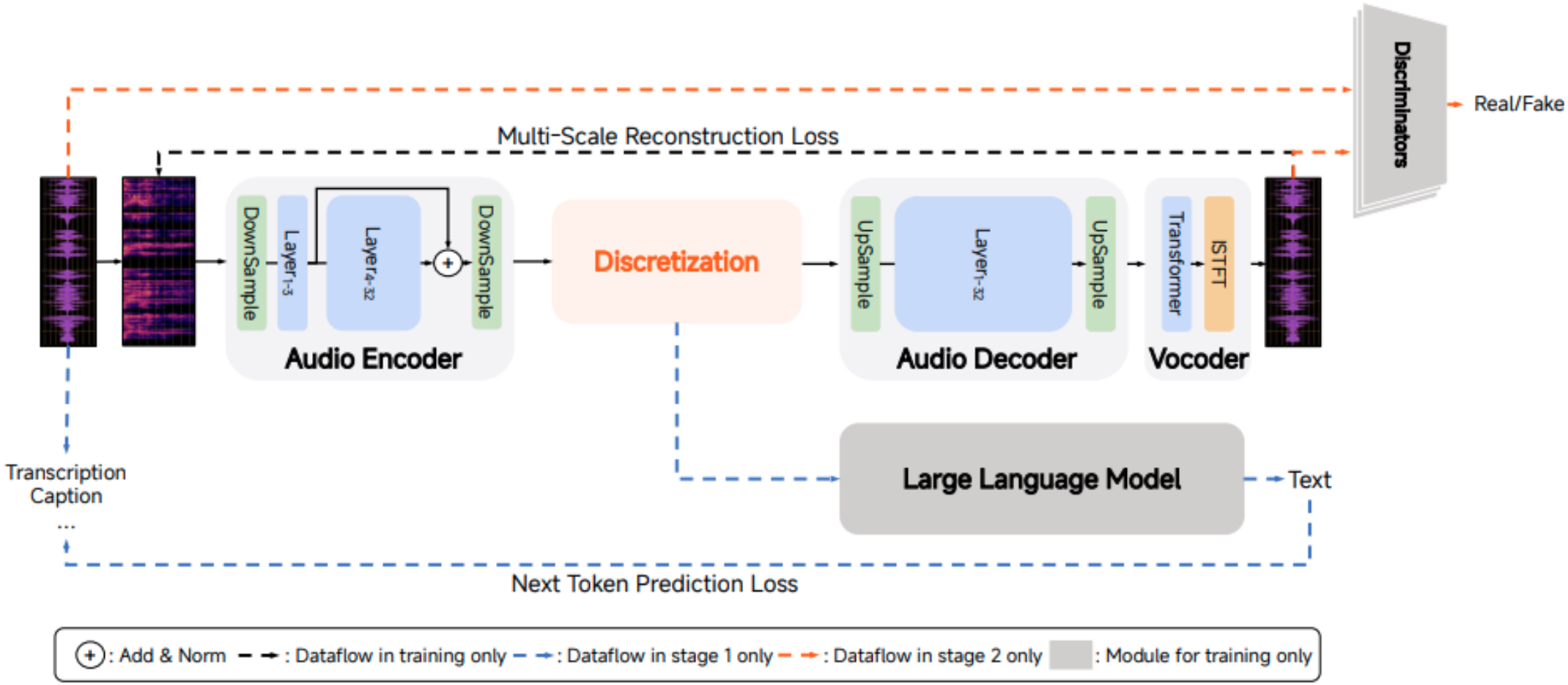



**Figure 2** Illustration of MiMo-Audio-Tokenizer framework.

Step-Audio 2 Technical Report 27 Aug 2025



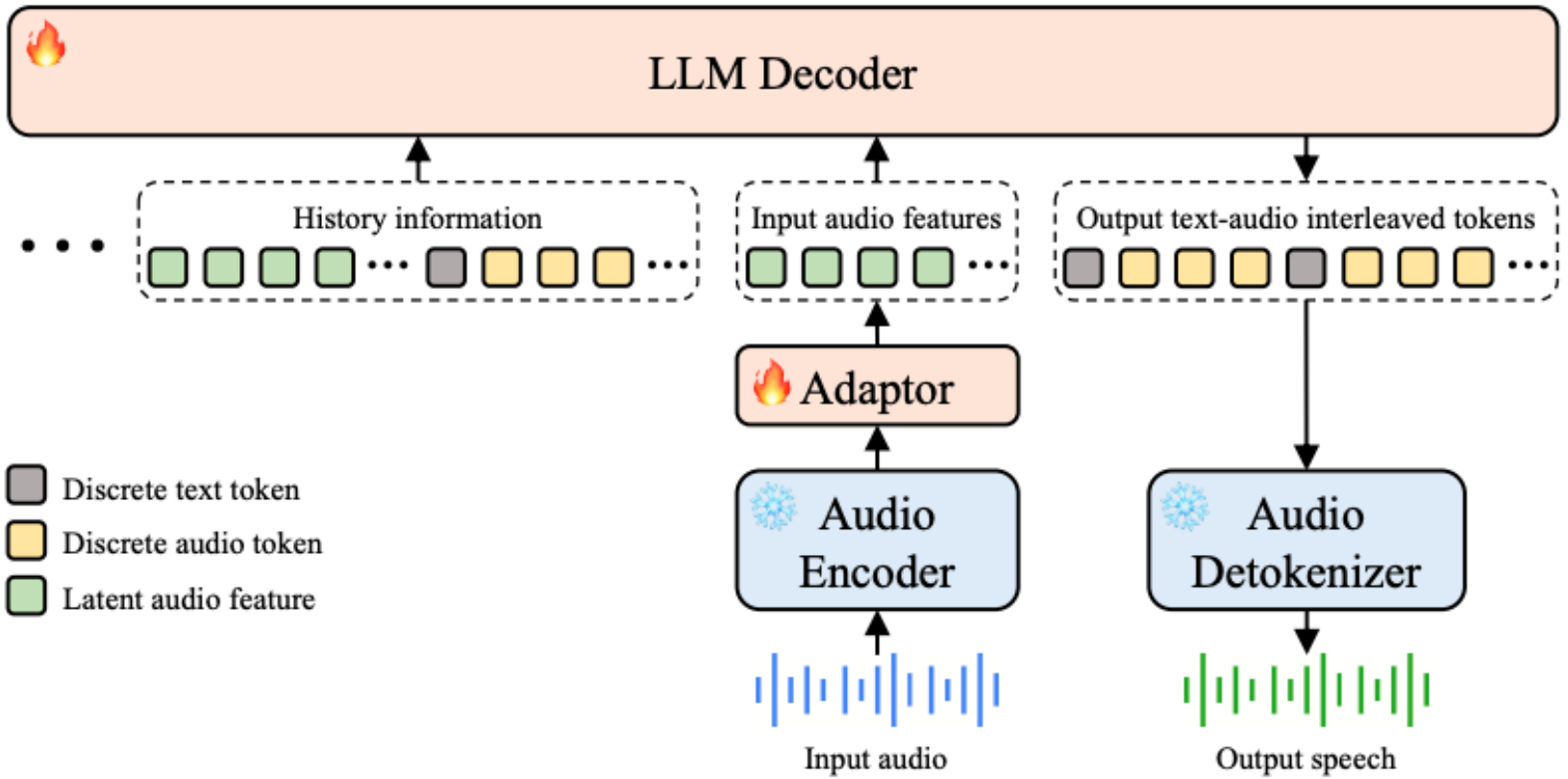



Figure 3: Architecture of the Step-Audio 2.

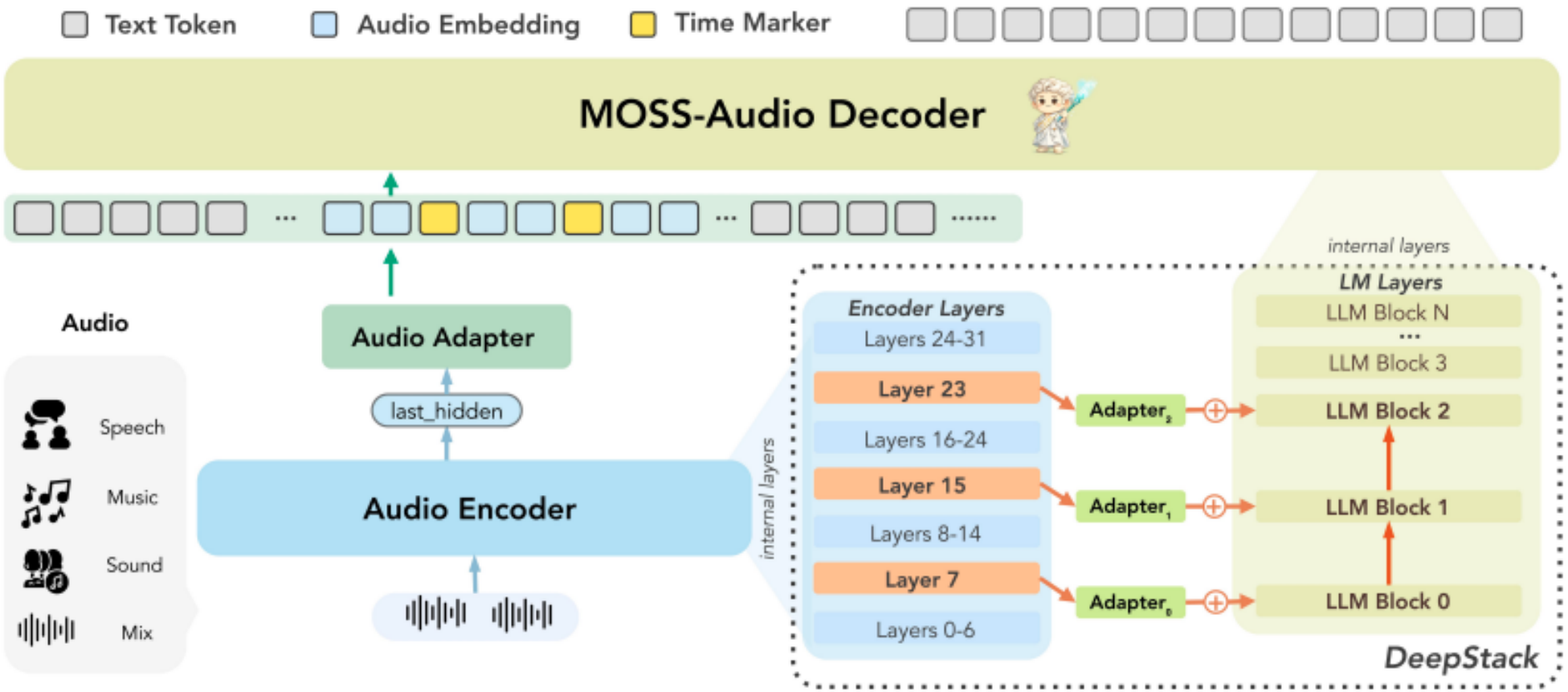



**Figure 2** Architecture of MOSS-Audio.

Step-Audio 2 Technical Report 27 Aug 2025



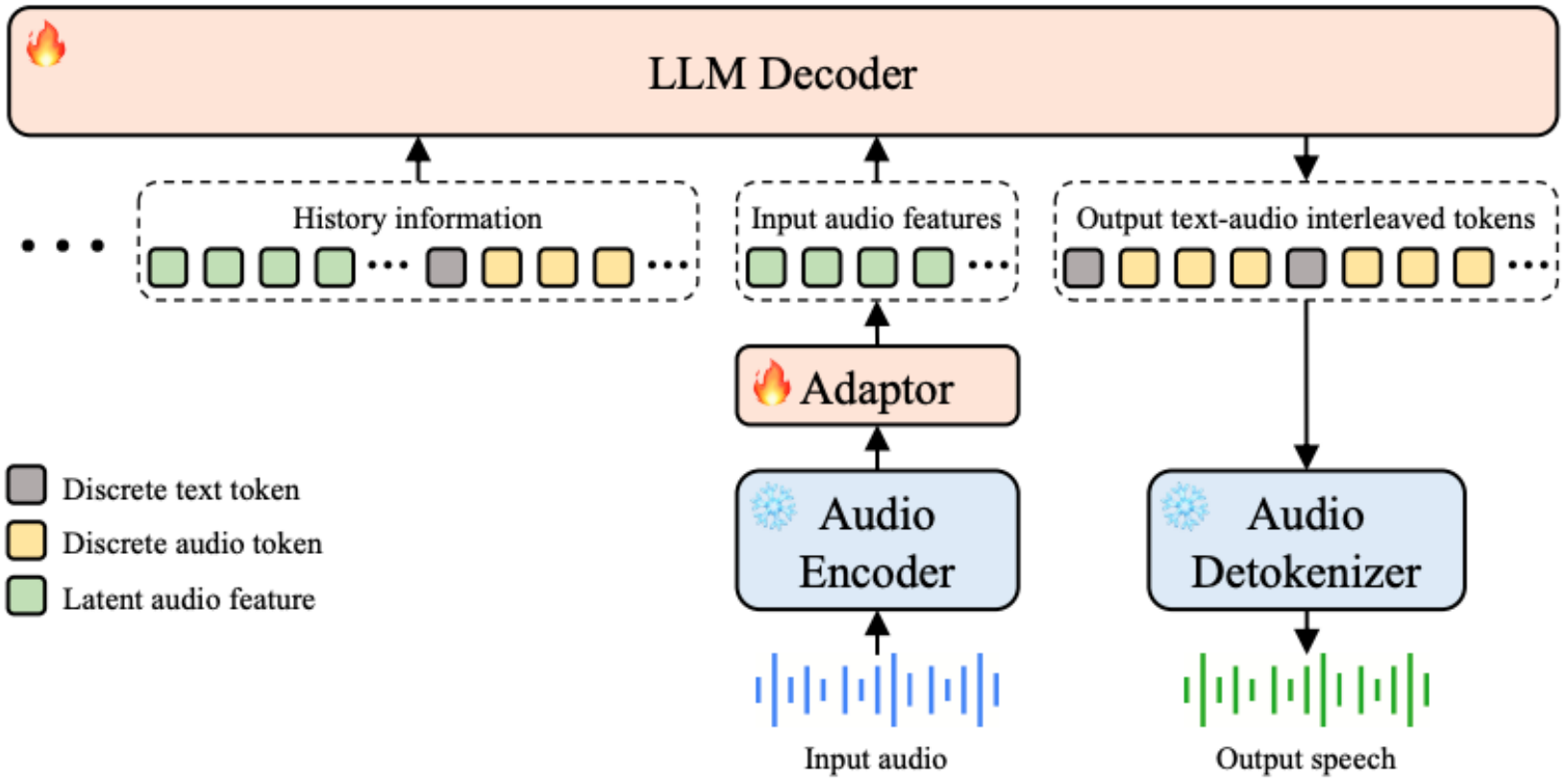



Figure 3: Architecture of the Step-Audio 2.

Qwen3-ASR Technical Report 2026-02-02

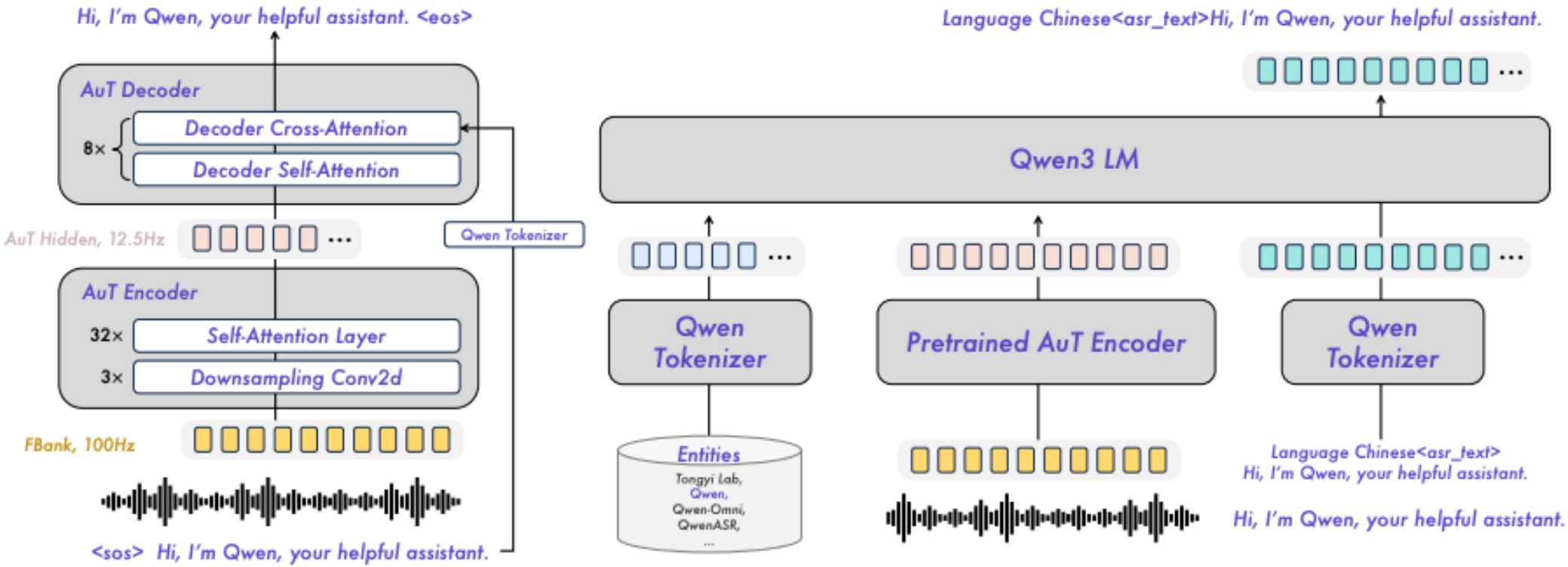



Figure 2: Architecture of AuT (left) and the overview of Qwen3-ASR (right).